

7(a)	Any five from: • (as temperature rises) energy of electrons increases • electrons (have enough energy to) cross forbidden band • electrons enter conduction band • leaving holes in valence band • both holes and electrons act as charge carriers • more charge carriers results in lower resistance • increased lattice vibrations outweighed by increase in (number of) charge carriers	B5
7(b)	(at 10 °C resistance is) 2.55 kΩ	C1
	new potential difference = $9.00 \times 2.55 / (2.55 + 12.0)$ = 1.58 V	C1
	change in p.d. = 0.58 V	A1
7(c)	change of resistance with temperature is not linear	B1
	change in potential with resistance is not linear or potential divider equation is non-linear	B1